

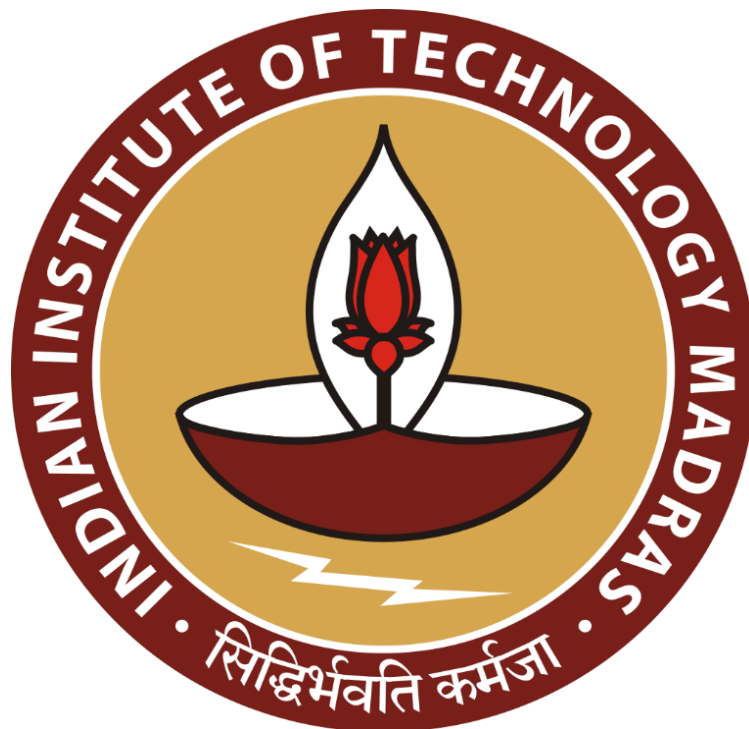
Overcoming Operational Bottlenecks In Laundry Services: A Case Of Uclean

A PROPOSAL REPORT FOR THE BDM CAPSTONE PROJECT

Submitted By

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Declaration Statement

I am working on a Project titled “**Overcoming Operational Bottlenecks in Laundry Services: A Case of UClean**”. I extend my appreciation to **UClean**, for providing the necessary resources that enabled me to conduct my project.

The utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully I hereby assert that the data presented and assessed in this project report is genuine and precise to analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfilment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.



Name: Aditi Saxena

Date: 10-05-2025

1. Executive Summary:

The project focuses on UClean's franchise located in Sector 37D, Gurgaon. UClean is India's largest and fastest growing laundry and dry-cleaning chain, founded by **Arunabh Sinha** in 2016. The UClean company is a B2C laundry and dry-cleaning service provider aiming to deliver high quality and efficient laundry solutions to customers. Operating on a franchise model, UClean serves residential and commercial clients, including hotels and offices, and offers subscription-based laundry services for added convenience.

Over four months, billing and expense data were analyzed to assess UClean's operations. Key issues identified include long turnaround times, frequent cancellations, and poor forecasting of seasonal demand, all leading to inefficiencies and reduced customer satisfaction. Additionally, lack of structured feedback analysis and unclear value communication make it difficult to retain cost-conscious customers in a competitive market.

Addressing these challenges is key to improving service efficiency, reducing delays, and enhancing customer retention. This project applies data-driven approaches like turnaround time tracking, demand forecasting, and feedback analysis. Microsoft Excel, Google Sheets, and Python (Matplotlib, Pyplot) will be used to analyze billing and operational data, visualize trends, and support informed decision-making for long-term improvements.

2. Organization Background:

UClean, as a leading laundry and dry-cleaning service provider operating in multiple locations, offering efficient cleaning solution to B2C customers. One of its franchises, located in Sector 37D, opposite to BPTP Generations, (Gurgaon) has recently completed its first year of operations. The UClean company operates on a franchise based model, allowing entrepreneurs to establish and run their own laundromats under the UClean brand.

This UClean's franchise, opened in January 2024 by **Mr. Ritu Raj**, a former **Salesforce Developer**, who transitioned into entrepreneurship to capitalize on the growing demand for professional laundry services. They serves both residential and commercial clients, including hotels, hospitals, and corporate offices with high-quality laundry and dry-cleaning services with a dedicated team of five staff members with one earning a salary of ₹25,000 and the remaining four earning ₹12,000 each.

The franchise store generates approximately ₹1.5 lakh in revenue per month but is currently operating at a loss. Over the past six months, it has catered to around 600-650 customers, averaging 100-110 customers per month.

3. Problem Statement:

3.1 Long Turnaround Time and Service Delays: Delays in laundry service caused by poor task distribution and limited staff directly impact customer satisfaction. When clothes aren't returned promptly especially for time sensitive needs it results in frustration, negative reviews, and loss of repeat customers.

3.2 High Order Cancellations and Operational Disruptions: Frequent last minute order cancellations due to delays or miscommunication. Each cancellation not only leads to a loss of revenue but also wastes valuable staff time and resources. They also reflect deeper service reliability issues that damage customer trust.

3.3 Inability to Forecast Seasonal Demand: The franchise sees spikes during festivals and events, but without forecasting tools, staff and resource planning remains reactive. This leads to overload during peaks and underutilization in off-peak periods, affecting efficiency and profits.

4. Background of the Problem:

UClean's franchise is encountering multiple operational challenges that are affecting its service delivery, customer retention, and overall financial performance. As a B2C laundry and dry-cleaning service, ensuring efficiency, reliability, and customer satisfaction is essential for sustaining profitability and brand loyalty.

4.1 Long Turnaround Time and Service Delays: The franchise often struggles with delayed order processing due to unbalanced workload distribution and limited manpower. These delays particularly affect customers with urgent needs, resulting in dissatisfaction, reduced repeat business, and negative feedback. Long turnaround times not only hurt service perception but also reduce the store's daily processing capacity.

4.2 High Order Cancellations and Operational Disruptions: Cancellations frequently occur when there are delays in pickups or lack of timely communication with customers. These disruptions lead to underutilized resources, inefficient labor usage, and revenue loss. Over time, they signal deeper reliability issues and contribute to declining customer trust and loyalty.

4.3 Inability to Forecast Seasonal Demand: UClean faces significant spikes in demand during weekends and festive seasons but lacks data-driven forecasting tools. As a result, staffing and machine usage become reactive rather than planned. This mismatch causes overload during peak periods and resource wastage in off-peak times, affecting both efficiency and profitability.

5. Problem Solving Approach:

5.1 Long Turnaround Time and Service Delays:

- **Workflow Optimization:** Service order timelines will be analyzed to identify delays at each stage (pickup, washing, ironing, delivery). Based on this, task reallocation and shift adjustments will be implemented to streamline workflows and reduce processing time.
- **Service Time Analysis:** Historical data will be used to calculate average service time per order and per employee. Employees will be trained to meet optimized turnaround benchmarks, and any significant deviations will be flagged for review.
- **Capacity Utilization Monitoring:** Machine usage logs will be reviewed to minimize idle time and plan maintenance during off-peak hours. This ensures that equipment is available when needed most.

5.2 Minimizing Cancellations with Predictive Alerts:

- **Cancellation Pattern Analysis:** Past cancellations will be categorized by time, service type, and customer type to identify common causes. This will inform staff training and allow for the redesign of pickup and delivery scheduling.
- **Flexible Time Slots:** Cancellation rates during specific hours will guide the restructuring of delivery windows. Offering flexible or buffer time slots may accommodate more customer needs and reduce service friction.

5.3 Forecasting Seasonal Demand with Time Series Models:

- **Time Series Forecasting:** Historical order data will be analyzed using time series methods (moving averages, ARIMA) to forecast demand during festivals, weekends, and school seasons. These predictions will guide pre-scheduling of additional staff and resources.
- **Service Prioritization Strategy:** During forecasted high-demand times, services will be categorized based on time-sensitivity. This ensures urgent orders are prioritized without compromising quality or overloading the system.

- **Dynamic Resource Allocation:** Based on forecasted data, staffing levels and delivery routes will be adjusted to align with expected demand. Staff hiring for temporary peak periods can also be scheduled in advance.

Data Analysis and Visualization:

Microsoft Excel and Google Sheets will be used to manage billing, expenses, and operational data. Tools like Pivot Tables, VLOOKUPs, charts, and graphs will help identify workload distribution, employee performance, and customer behavior trends.

To visualize complex data relationships, Python libraries such as Matplotlib and Pyplot will be used. These graphical tools will highlight service bottlenecks, seasonal patterns, and customer churn, enabling informed and data-driven decision-making.

6. Expected Timeline:

6.1 Work Breakdown Structure:

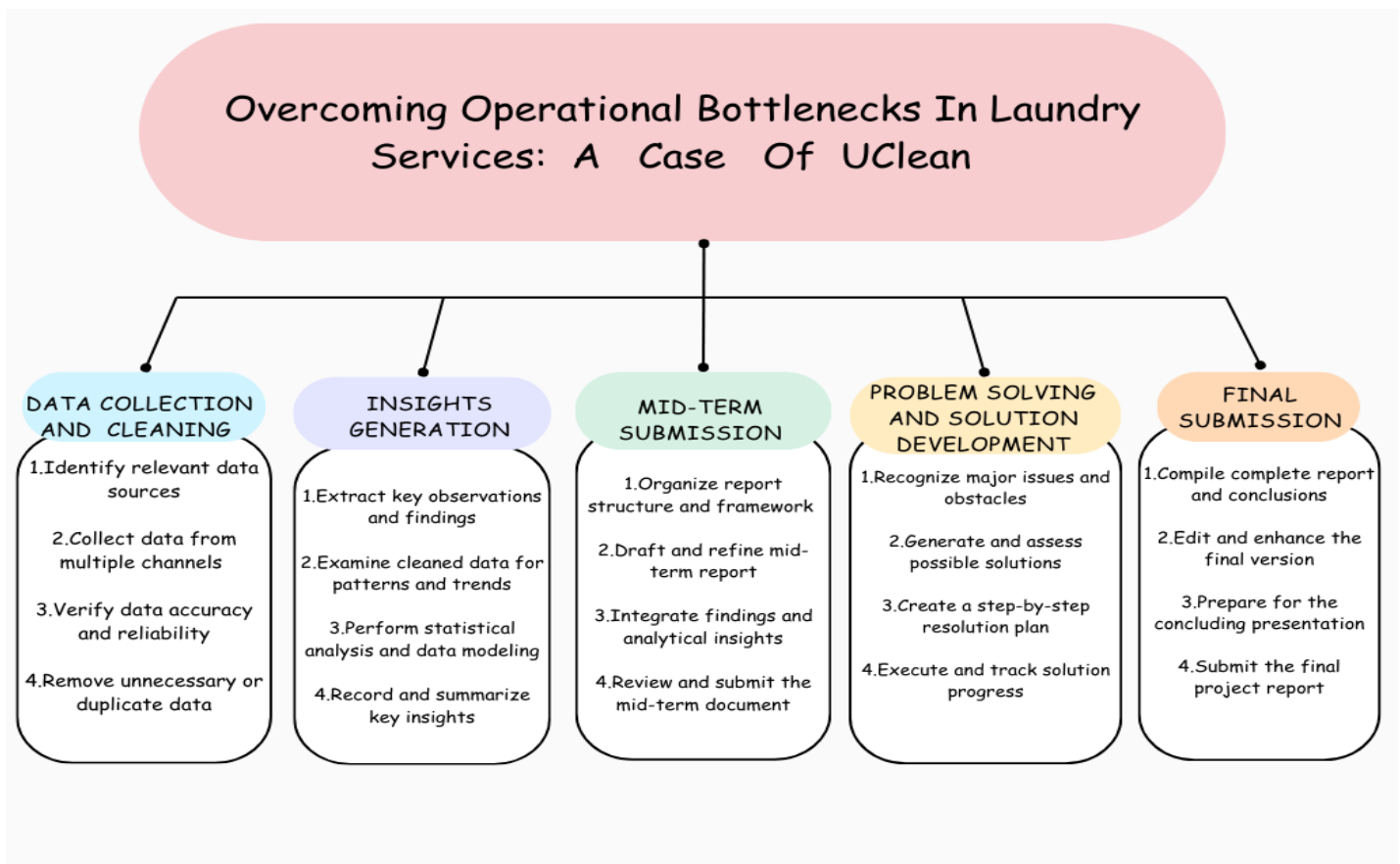


Figure 1. Work Breakdown Structure for the project.

6.2 Gantt Chart:

Task	Start Date	End Date
BDM Project		
Meeting 1	10-05-2025	15-05-2025
Meeting 2	20-05-2025	22-05-2025
Data Collection Round 1	25-05-2025	26-05-2025
Proposal Submission	11-06-2025	13-06-2025
Data Collection Round 2	14-06-2025	17-06-2025
Data Pre-Processing	17-06-2025	19-06-2025
Data Analysis	20-06-2025	25-06-2025
Mid Term Submission	06-07-2025	08-07-2025
Final Presentation Submission	01-08-2025	03-08-2025

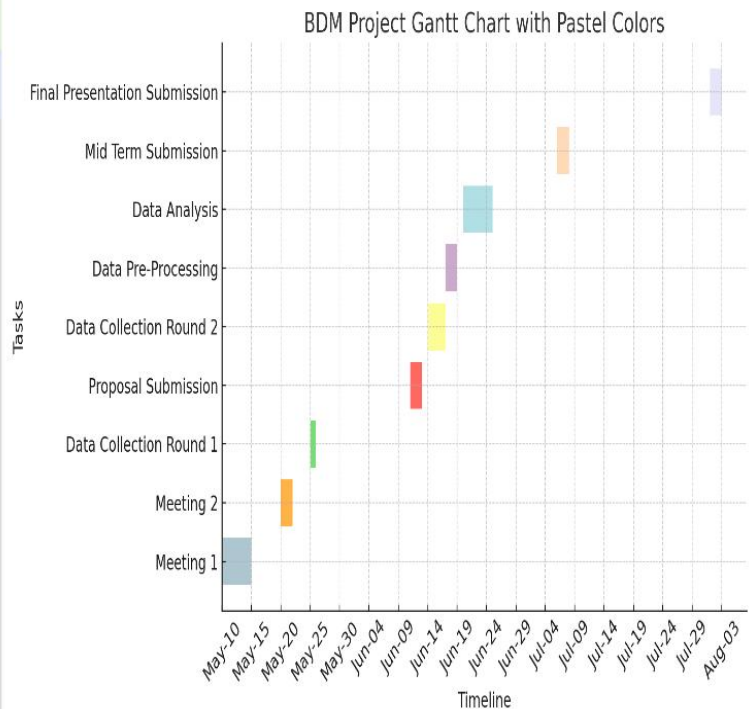


Figure 2. Expected Timeline for Completion of Project.

7. Expected Outcome:

The proposed data-driven solutions are expected to enhance operational efficiency and customer satisfaction at UClean's franchise. Streamlined task allocation and better scheduling will reduce turnaround time and delays. Predictive tools will lower order cancellations, while demand forecasting will improve resource planning during peak seasons.

Customer feedback analysis will help identify and resolve key pain points, leading to improved service quality. Value based pricing and loyalty strategies will boost retention and revenue. Overall, the franchise will become more responsive, reliable, and profitable.